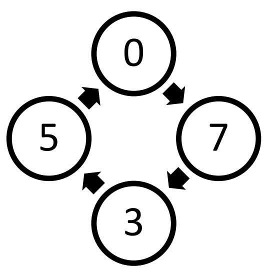
MID-TERM EXAMINATION

Course: Introduction to Computer Hardware (20CTT1-A)	
Time: 75 minutes	Term: 3 – Academic year: 2021-2022
Lecturer(s): **Dinh Dien**	
Student name:	Student ID:

(Notes: Neither books nor laptops, phones allowed)

1. Design a circuit that will tell whether a given month has 31 days in it. The month is specified by a 4-bit input A3:0. For example, if the inputs are 0001, the month is January, and if the inputs are 1100, the month is December. The circuit output Y should be HIGH only when the month specified by the inputs has 31 days in it. Write the simplified equation, and draw the circuit diagram using a minimum number of gates.
2. A multiplier circuit employing Booth algorithm is used to multiply two numbers +9 (multiplican) and -5 (multiplier) expressed in 2’s complement binary notation. Show the Booth recording for the multiplier and calculate the product by using Booth algorithm.
3. Design a to Led7Seg decoder using common Anode Led7seg for binay code 10, 11, 12, 13, 14, 15 to E, F, C, H, L, d. Don’t care binary code from 0 to 9.
4. Design a synchronous up counter using JK-FlipFlops. The value of counter is shown in following diagram



Note: All problems need explain in your design

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